Network Effects in the v8 SAOM Specification

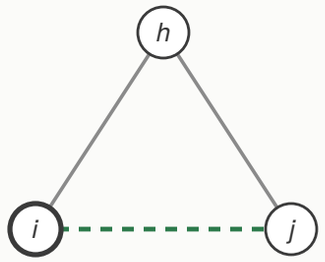
Each panel shows the collaboration tie under consideration (dashed) from the focal inventor *i*. All ties are undirected co-inventing relations.

incumbent · newcomer · existing tie · tie being modeled · prior career ties (covariate)

1. Density (outdegree)

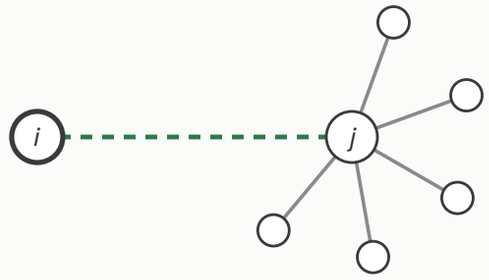
Baseline tendency to form any tie: the "cost" of collaboration. Anchors the model; nearly always strongly negative.

2. Transitive ties (transTies)



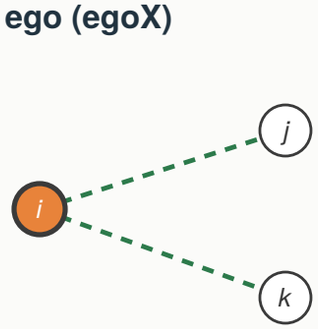
Closure: a shared partner h pulls the i–j tie shut. Collaborators of my collaborators become mine.

3. Degree activity + popularity (degPlus)



Preferential attachment: ties gravitate toward inventors who are already well connected in this network.

4. Newcomer ego (egoX)



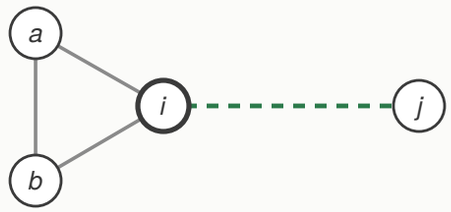
Do newcomers form more (or fewer) ties overall than incumbent inventors?

5. Career tenure ego (egoX)

years of experience

Does experience change tie formation? (Unstable when tenure barely varies; the Case C parameter.)

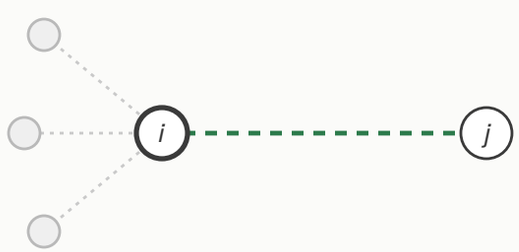
6. Network constraint ego (egoX)



closed cluster (high constraint)

Does embeddedness in a closed cluster (Burt's constraint) dampen reaching out to new partners?

7. Inventor degree ego (egoX)

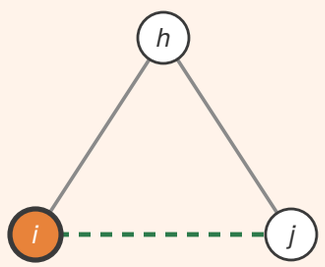


career collaborations (covariate)

Does prior collaboration breadth across the career (not this network) drive new tie formation?

8. Newcomer × transitive ties

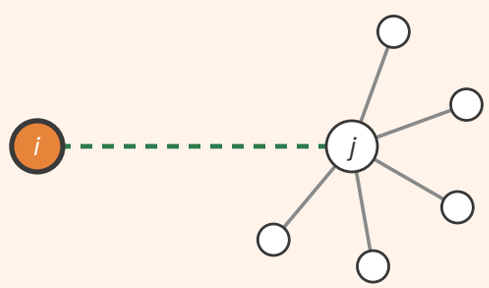
theory-bearing, new in v8



Do newcomers close triads at different rates than incumbents? Pooled estimate: negative (μ = –1.95).

9. Newcomer × degPlus

theory-bearing, new in v8



Do newcomers attach to hubs differently than incumbents? Pooled estimate: null (μ ≈ 0).

v8 specification; RSiena effect names in parentheses. The rate function is estimated per network but not shown. Panels 8–9 interact the newcomer indicator with the structural effects in panels 2–3.